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Structure of transferring dies for use in mass transferring process

Abstract

A structure of transferring dies includes an oxide layer supporting feature, multiple dies, a bonding feature, a supporting wafer, and a spacer. The oxide layer supporting feature includes multiple repeating units. Each repeating unit has a die setting region and a peripheral region. The die setting region of one repeating unit is separated from the peripheral region of another adjacent repeating unit. The die is disposed on the die setting region and the bonding feature is disposed on the peripheral region of the oxide layer supporting feature. The supporting wafer is disposed under the oxide layer supporting feature and separated from the die and the bonding feature by a gap. The spacer is disposed between the bonding feature and the supporting wafer, and bonded to the bonding feature.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present disclosure relates generally to the technology of mass transfer of dies, and more particularly to structures and methods of transferring dies used in the process of mass transferring dies, and die bonded structures after the dies are transferred.

2. Description of the Prior Art

(2) As the development of technology, dies have been widely used in various electronic devices. A mass transfer method is usually used to place a large number of dies on the substrate of an electronic device for subsequent assembly. The method of placing the dies on the substrate of the electronic device is, for example, surface mount technology (SMT), wafer-to-wafer transfer technology, electrostatic transfer technology, elastomer stamp micro-transfer-printing technology, etc., where the elastomer stamp micro-transfer-printing technology uses an elastomer stamp as a pick-up head, fine-tunes the speed and force of the pick-up head, and destroys a tether structure of

the component to achieve the action of picking up the dies, which can transfer a large number of dies to the substrate of the electronic device.

(3) The tether structure of the component needs to be connected to a suspended die. Generally, the die is suspended by removing the material layer under the die through an etching process. However, the etching process is not easy to be controlled, and there may be some problems of insufficient etching or excessive etching to cause the efficiency of mass transfer of dies to be lower, and the number, the size, and the arrangement of transferred dies are limited.

SUMMARY OF THE INVENTION

(4) In view of this, embodiments of the present disclosure provide improved structures and methods of transferring dies to enhance the efficiency of transferring dies, increase the number of transferred dies per unit area, and improve the tolerance of the size and the arrangement of the dies.

(5) According to one embodiment of the present disclosure, a structure of transferring dies is provided and includes an oxide layer supporting feature, a plurality of dies, a bonding feature, a supporting wafer, and a spacer. The oxide layer supporting feature includes a plurality of repeating units, where each of the repeating units includes a die setting region and a peripheral region, and the die setting region of one repeating unit is separated from the peripheral region of another adjacent repeating unit. The dies are disposed on the die setting regions of the oxide layer supporting feature. The bonding feature is disposed on the peripheral region of the oxide layer supporting feature. The supporting wafer is disposed under the oxide layer supporting feature and separated from the die and the bonding feature by a gap. The spacer is disposed between the bonding feature and the supporting wafer, where the spacer is bonded to the bonding feature.

(6) According to one embodiment of the present disclosure, a method of transferring dies is provided and includes the following steps. A semiconductor-on-insulator (SOI) substrate is provided and includes a semiconductor substrate, an oxide layer and a semiconductor layer, where the oxide layer is disposed between the semiconductor substrate and the semiconductor layer. A plurality of dies and a bonding feature are formed on the oxide layer, where the bonding feature is separated from the dies, the die is disposed on a die setting region of a repeating unit of the oxide layer, and the bonding feature is disposed on a peripheral region of the repeating unit of the oxide layer. A supporting wafer is provided and a spacer is formed on the supporting wafer. A bonding process is performed to bond the supporting wafer to the bonding feature through the spacer, where the supporting wafer is separated from the die and the bonding feature by a gap. An etching process is performed on the oxide layer to separate the die setting region of one repeating unit from the peripheral region of another adjacent repeating unit of the oxide layer. In addition, the semiconductor substrate of the SOI substrate is removed to expose the oxide layer after the bonding process.

(7) According to one embodiment of the present disclosure, a die bonded structure is provided and includes a carrier substrate, a plurality of dies, a solder feature, and an oxide layer. The dies are bonded on the carrier substrate. The die has a first surface and a second surface opposite to the first surface, where the second surface is away from the carrier substrate. The solder feature is disposed between the first surface of the die and the carrier substrate. The oxide layer is disposed on the second surface of the die, where a portion of the oxide layer laterally protrudes from the die.

(8) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description

when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features may not be drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 shows schematic cross-sectional diagrams of two structures of transferring dies having different spacers and different supporting wafers according to some embodiments of the present disclosure.

(3) FIG. 2 shows schematic cross-sectional diagrams of two structures of transferring dies having different spacers and different supporting wafers according to some other embodiments of the present disclosure.

(4) FIG. 3 is a top view of a structure of transferring dies according to some embodiments of the present disclosure.

(5) FIG. 4 shows schematic cross-sectional diagrams of structures at several stages of a method of transferring dies and a die bonded structure after the dies are transferred according to one embodiment of the present disclosure.

(6) FIG. 5 shows schematic cross-sectional diagrams of structures at some stages of a method of transferring dies according to another embodiment of the present disclosure.

(7) FIG. 6 shows schematic cross-sectional diagrams of structures at several stages of a method of transferring dies and a die bonded structure after the dies are transferred according to another embodiment of the present disclosure.

(8) FIG. 7 shows schematic cross-sectional diagrams of structures at some stages of a method of transferring dies according to another embodiment of the present disclosure.

(9) FIG. 8 shows schematic cross-sectional diagrams of structures at some stages of a method of transferring dies and a die bonded structure after the dies are transferred according to another embodiment of the present disclosure.

(10) FIG. 9 is a top view of a die bonded structure after the dies are transferred according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

(11) The following disclosure provides many different embodiments, or examples, for implementing different features of the disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(12) Further, spatially relative terms, such as “beneath”, “below”, “lower”, “under”, “on”, “over”, “above”, “upper”, “bottom”, “top” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element (s) or feature (s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” and/or “under” other elements or features would then be oriented “above” and/or “over” the other elements or features. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(13) It is understood that, although the terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used

to distinguish one element, component, region, layer and/or section from another region, layer and/or section. Terms such as “first”, “second”, and other numerical terms when used herein do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer and/or section discussed below could be termed a second element, component, region, layer and/or section without departing from the teachings of the embodiments.

(14) As disclosed herein, the term “about” or “substantial” generally means within 20%, 10%, 5%, 3%, 2%, 1%, or 0.5% of a given value or range. Unless otherwise expressly specified, all of the numerical ranges, amounts, values and percentages disclosed herein should be understood as modified in all instances by the term “about” or “substantial”. Accordingly, unless indicated to the contrary, the numerical parameters set forth in the present disclosure and attached claims are approximations that may vary as desired.

(15) Although the disclosure is described with respect to specific embodiments, the principles of the invention, as defined by the claims appended herein, can obviously be applied beyond the specifically described embodiments of the invention described herein. Moreover, in the description of the present disclosure, certain details have been left out in order to not obscure the inventive aspects of the disclosure. The details left out are within the knowledge of a person having ordinary skill in the art.

(16) The present disclosure is directed to structures and methods for mass transfer (or referred to as mega transfer) of dies, where an oxide layer in a semiconductor-on-insulator (SOI) substrate is used to form a tether structure with anchor in the structure of transferring dies. According to embodiments of the present disclosure, while forming the tether structure with anchor for transferring dies, it is not necessary to etch the material layer under the dies, such that there is no problem of insufficient etching or excessive etching. Meanwhile, the tolerance of the size and the arrangement of the transferred dies are improved, thereby increasing the number of transferred dies per unit area, and improving the efficiency of transferring dies. In addition, according to embodiments of the present disclosure, in a die bonded structure after the dies are transferred, there is an oxide layer on the backside of the die, which protects the die and provides the support for the die with reduced thickness, thereby increasing the mechanical strength of the die bonded structure.

(17) FIG. 1 shows schematic cross-sectional diagrams of two structures of transferring dies according to some embodiments of the present disclosure, where structures **100-1** and **100-2** have different spacers and different supporting wafers. Please refer to the structure **100-1** of FIG. 1, which is a schematic cross-sectional diagram of a structure **100** of transferring dies according to one embodiment of the present disclosure. The structure **100** of transferring dies is an intermediate structure used in the process of mass transfer of dies. In order to make the diagrams concise and easy to understand, all the cross-sectional diagrams show one die. In fact, the number of dies to be transferred may be in the thousands or millions. The transferred dies may be applied to, for example, micro light emitting diode (μ -LED) display devices as driving chips for μ -LED, but not limited thereto. As shown in the structure **100-1** of FIG. 1, according to one embodiment of the present disclosure, the structure **100-1** of transferring dies includes an oxide layer supporting feature **101** that has a plurality of repeating units **101U**. Each repeating unit **101U** has a die setting region **101A** and a peripheral region **101B**, and the die setting region **101A** of one repeating unit **101U** is separated from the peripheral region **101B** of another adjacent repeating unit. For each repeating unit **101U**, a tether structure like a tether state is disposed at the interface between the die setting region **101A** and the peripheral region **101B**. Through the tether structure, the die setting region **101A** is connected to the peripheral region **101B**. A plurality of dies **107** is respectively disposed on the die setting regions **101A** of the oxide layer supporting feature **101**, and a bonding feature **109** is disposed on the peripheral region **101B** of the oxide layer supporting feature **101**. According to embodiments of the present disclosure, the oxide layer supporting feature **101** may be formed from a buried oxide layer in a semiconductor-on-insulator (SOI) substrate, and the material of the oxide layer supporting feature **101** is, for example, silicon oxide. Both the die **107** and the

bonding feature **109** may be formed from a semiconductor layer **103** and an interconnect structure layer **105** on the semiconductor layer **103** of the SOI substrate, such that the die **107** and the bonding feature **109** have substantially the same semiconductor layer **103** and the same interconnect structure layer **105**, and the die **107** and the bonding feature **109** are separated from each other. The semiconductor layer **103** includes silicon, polysilicon or other semiconductor materials. The interconnect structure layer **105** includes multiple dielectric layers and multiple conductive layers. The interconnect structure layer **105** is electrically connected to various elements in the semiconductor layer **103**. According to an embodiment of the present disclosure, dummy elements are disposed in the semiconductor layer **103** of the bonding feature **109**, and dummy interconnections are disposed in the interconnect structure layer **105** of the bonding feature **109**.

(18) In one embodiment, the die **107** has a plurality of conductive pads **115**, and the bonding feature **109** also has a bonding pad **117**. A passivation layer **116** may be disposed on the surfaces of the die **107** and the bonding feature **109**. The passivation layer **116** has multiple openings to expose the conductive pads **115** of the die **107** and the bonding pad **117** of the bonding feature **109**. In addition, the structure **100-1** of transferring dies further includes a protective layer **120** that conformally covers the sides and the front surface of the die **107**, the sides and the front surface of the bonding feature **109**, and the exposed surface of the oxide layer supporting feature **101** between the die **107** and the bonding features **109**. The protective layer **120** has a plurality of openings that respectively expose the conductive pads **115** of the die **107** and the bonding pad **117** of the bonding feature **109**. In some embodiments, the material of the protective layer **120** includes silicon oxide, silicon nitride, silicon oxynitride, a combination thereof, or other insulating materials. According to an embodiment of the present disclosure, the thickness of the oxide layer supporting feature **101** in the die setting region **101A** is between about 2 μm and about 4 μm (referred to as a first thickness). The sum of the thickness of the oxide layer supporting feature **101** in the die setting region **101A** and the thicknesses of the die **107** is less than about 15 μm (referred to as a second thickness). Thus, the ratio of the first thickness to the second thickness is between 0.2 and 1.0.

(19) As shown in FIG. 1, according to an embodiment of the present disclosure, the structure **100-1** of transferring dies further includes a supporting wafer **111** disposed under the oxide layer supporting feature **101**, where the supporting wafer **111** is separated from the die **107** and the bonding feature **109** by a gap. In addition, the structure **100-1** of transferring dies further includes a spacer **113** disposed between the bonding feature **109** and the supporting wafer **111**, where the spacer **113** is bonded to the bonding pad **117** of the bonding feature **109**. In this embodiment, the spacer **113** includes a glue bond dam to perform bonding through adhesive force.

(20) According to embodiments of the present disclosure, during the process of transferring dies, the oxide layer supporting feature **101** in the structure **100-1** of transferring dies supports the die **107** and the bonding feature **109** that are separated from each other, and after a die transfer equipment is attached to the oxide layer supporting feature **101**, a portion of the oxide layer supporting feature **101** between the die **107** and the bonding feature **109** is used as a tether structure. When the die transfer equipment applies torsion force on the oxide layer supporting feature **101**, the portion of the oxide layer supporting feature **101** as the tether structure will be broken to make the die **107** to be released from the structure **100-1** of transferring dies, and then the die **107** is transferred and bonded onto a carrier substrate of an electronic device. In addition, the spacer **113** in the structure **100-1** of transferring dies keeps a gap between the die **107** and the supporting wafer **111** to prevent the die **107** from being damaged by collision during the process of transferring dies. In addition, the supporting wafer **111** provides more support to the die **107** having a reduced thickness during the process of transferring dies, thereby strengthening the overall mechanical strength of the structure **100-1** of transferring dies.

(21) Please refer to the structure **100-2** of FIG. 1, which is a schematic cross-sectional diagram of a structure **100** of transferring dies according to another embodiment of the present disclosure. The

difference between the structure **100-2** and the structure **100-1** of FIG. 1 is that the supporting wafer **111** of the structure **100-2** of transferring dies has a cavity **112** corresponding to the position of the die **107**, and the distance **d1** between the die **107** and the bottom surface of the cavity **112** of the supporting wafer **111** is greater than the distance **d2** between the bonding feature **109** and the surface of the supporting wafer **111**. In addition, a spacer **114** of the structure **100-2** of transferring dies includes a eutectic bond bump **114a** and a stand-off **114b**, where the stand-off **114b** may be formed by etching a portion of the supporting wafer **111**, i.e., the stand-off **114b** and the supporting wafer **111** may be integrally formed. In this embodiment, bonding is performed by using the eutectic bond bump **114a** to produce eutectic bonding with the bonding pad **117** of the bonding feature **109**. The other components of the structure **100-2** of transferring dies may refer to the aforementioned descriptions of the structure **100-1**, and will not be repeated here.

(22) FIG. 2 shows schematic cross-sectional diagrams of two structures of transferring dies according to some other embodiments of the present disclosure, where structures **100-3** and **100-4** of transferring dies have different spacers and different supporting wafers. A spacer **113** and a supporting wafer **111** in the structure **100-3** of transferring dies of FIG. 2 may refer to the aforementioned description of the structure **100-1** of FIG. 1, and will not be repeated here. A space **114** and a supporting wafer **111** having a cavity **112** in the structure **100-4** of transferring dies of FIG. 2 may refer to the aforementioned description of the structure **100-2** of FIG. 1, and will not be repeated here. The difference between the embodiments of FIG. 2 and the embodiments of FIG. 1 is that in the structures **100-3** and **100-4** of transferring dies of FIG. 2, an under bump metallurgy (UBM) **122** and a metal bump **124** are further disposed on the conductive pad **115** of the die **107**. The UBM **122** is conformally formed on the surface of the conductive pad **115** exposed by the opening of the passivation layer **116**, on the sidewalls of the opening of the passivation layer **116**, and on a portion of the surface of the passivation layer **116**. The metal bump **124** is disposed on the UBM **122**. The metal bump **124** fills up the opening of the passivation layer **116** and extends to a portion of the surface of the passivation layer **116**. In addition, in the embodiments of FIG. 2, the protective layer **120** further covers the sidewalls of the UBM **122** and the metal bump **124**, and also extend to a portion of the surface of the metal bump **124**. The opening of the protective layer **120** exposes another portion of the surface of the metal bump **124**.

(23) FIG. 3 is a top view of a structure of transferring dies according to some embodiments of the present disclosure, where the area A corresponds to the structures of FIG. 1 and FIG. 2. In order to make the diagram concise and easy to understand, FIG. 3 only depicts the oxide layer supporting feature **101**, the spacer **113/114**, and the supporting wafer **111**. As shown in FIG. 3, in some embodiments, the width (e.g., in the Y direction) of the oxide layer supporting feature **101** at the interface between the die setting region **101A** and the peripheral region **101B** is narrowed, such that the oxide layer supporting feature **101** is used as a tether structure for transferring dies. When the die transfer equipment applies a torsion force on the oxide layer supporting feature **101**, the narrowed portion will be broken, so that the die is released from the structure **100** of transferring dies. In some embodiments, the spacer **113/114** may be an elongated structure disposed corresponding to the peripheral region **101B**. In other embodiments, two adjacent spacers **113/114** may also construct a portion of a ring structure. In addition, the die setting region **101A** of the oxide layer supporting feature **101** may correspond to the position of placing the die.

(24) According to embodiments of the present disclosure, since the tether structure with an anchor for transferring dies is provided by the oxide layer supporting feature **101**, and the supporting wafer **111**, the spacer **113/114** and the bonding feature **109** make the die **107** to be suspended, the dies are transferred without etching the material layer under the dies **107**. Therefore, the size of the die, the layout density of the dies, the distance between the dies (such as the distance **S1** between the die setting regions **101A** as shown in FIG. 3), and the distance **S2** between the die setting region **101A** of one repeating unit and the peripheral region **101B** of another adjacent repeating unit (such as the distance **S2** as shown in FIG. 3) are not limited by an etching process. The aforementioned

distances **S1** and **S2** may be reduced, and the layout density of the dies may be increased, so that the number of transferred dies per unit area is increased, and the efficiency of transferring dies is improved.

(25) FIG. 4 shows schematic cross-sectional diagrams of structures at several stages of a method of transferring dies and a die bonded structure after the dies are transferred according to one embodiment of the present disclosure. As shown in FIG. 4, in one embodiment, first, a SOI substrate **110** is provided. The SOI substrate **110** includes a semiconductor substrate **102**, an oxide layer **104** and a semiconductor layer **103**, where the oxide layer **104** is disposed between the semiconductor substrate **102** and the semiconductor layer **103**. A plurality of electronic devices such as transistors, resistors, capacitors, etc. are formed in the semiconductor layer **103**, and an interconnection structure layer **105** is formed on the semiconductor layer **103**. The conductive pads **115** of the die **107** and the bonding pad **117** of the bonding feature **109** may be formed from the topmost conductive layer of the interconnect structure layer **105**. The passivation layer **116** is formed on the interconnect structure layer **105**, and has a plurality of openings to respectively expose the conductive pads **115** and the bonding pad **117**. Next, a patterning process is performed on the interconnect structure layer **105** and the semiconductor layer **103** to form a plurality of dies **107** and bonding features **109** on the oxide layer **104**, where the bonding feature **109** is separated from the die **107**, which may refer to the aforementioned description of FIG. 1. The die **107** is disposed on the die setting region **101A** of the repeating unit of the oxide layer **104**, and the bonding feature **109** is disposed on the peripheral region **101B** of the repeating unit of the oxide layer **104**. Then, a protective layer **120** is deposited to conformally cover the sides and the front surface of the die **107**, the sides and the front surface of the bonding feature **109**, and the exposed surface of the oxide layer **104** between the die **107** and the bonding feature **109**. The protective layer **120** is etched to form a plurality of openings to respectively expose the conductive pads **115** of the die **107** and the bonding pad **117** of the bonding feature **109**.

(26) Next, as shown in FIG. 4, at step **S401**, an etching process **410** is performed on the protective layer **120** and the oxide layer **104**, so that the protective layer **120** and the oxide layer **104** in the die setting region **101A** of one repeating unit are separated from the protective layer **120** and the oxide layer **104** in the peripheral region **101B** of another adjacent repeating unit, and an oxide layer supporting feature **101** is formed.

(27) Then, as shown in FIG. 4, at step **S403**, a supporting wafer **111** is provided, and spacers **113** are formed on the supporting wafer **111**. A bonding process **420** is performed to bond the supporting wafer **111** to the bonding pads **117** of the bonding features **109** through the spacers **113**, where the supporting wafer **111** is separated from the die **107** and the bonding features **109** by a gap. In this embodiment, the bonding process **420** may be performed through the glue of the spacer **113**.

(28) Afterwards, as shown in FIG. 4, at step **S405**, the structure of the step **S403** is turned upside down, and the semiconductor substrate **102** of the SOI substrate **110** may be completely removed to expose the oxide layer supporting feature **101**, and the structure **100-1** of transferring dies as shown in the FIG. 1 is formed.

(29) Next, as shown in FIG. 4, at step **S407**, a die transfer equipment (hereinafter may be referred to as a transfer equipment) **330** is provided, which is temporarily fixed on the die setting region **101A** of the oxide layer supporting feature **101** through an attachment feature **332**. In some embodiments, the attachment feature **332** includes an adhesive, an UV-curable tape, an adhesive fixture using Van der Waals force (such as an adhesive stamp), or a mechanical clamp, so that the transfer equipment **330** adheres or adsorbs on the oxide layer supporting feature **101**, or the transfer equipment **330** may directly clamp the die **107** and the oxide layer supporting feature **101** by using the mechanical clamp.

(30) Then, as shown in FIG. 4, still at the step **S407**, a torsion force **334** is applied by the transfer equipment **330** to fracture the tether structure of the oxide layer supporting feature **101** at the

interface between the die setting region **101A** and the peripheral region **101B** in the same repeating unit, so that the die setting region **101A** and the peripheral region **101B** of the same repeating unit are separated from each other. As a result, at step **S409**, the die **107** is released from the structure **100-1** of transferring dies, and a portion of the oxide layer supporting feature **101** (such as the die setting region **101A**) is remained on the die **107** and attached to the transfer equipment **330**.

(31) Thereafter, as shown in FIG. 4, at step **S411**, the plurality of dies **107** and the portion of the oxide layer supporting feature **101** (such as the die setting region **101A**) remained on the backside surface of the die **107** are transferred by the transfer equipment **330** onto a carrier substrate **201**. After the plurality of dies **107** are transferred onto the carrier substrate **201**, the transfer equipment **330** and the attachment feature **332** are detached from the remained portion of the oxide layer supporting feature **101** to complete a die bonded structure **200** as shown in FIG. 4. In some embodiments, the carrier substrate **201** includes a glass substrate, a flexible plastic substrate, a printed circuit board, or other device substrates of applicable electronic products. In addition, the plurality of dies **107** may be bonded to the carrier substrate **201** through solder features **202**. The solder features **202** are disposed between the carrier substrate **201** and a first surface of the die **107** facing the carrier substrate **201**. In some embodiments, the solder features **202** include solder balls or solder bumps, and the solder features **202** are bonded to the conductive pads **115** of the die **107**.

(32) As shown in FIG. 4, according to some embodiments of the present disclosure, after the dies are transferred, the die bonded structure **200** includes a portion of the oxide layer supporting feature **101** (such as the die setting region **101A**, which may also be referred to as an oxide layer) disposed on a second surface of the die **107** away from the carrier substrate **201**. In addition, while viewed in a top view, a portion of the oxide layer laterally protrudes from the die **107**, so that the area of the oxide layer is larger than the area of the die **107** in a top view. Moreover, the die bonded structure **200** further includes a protective layer **120** covering the side surfaces and the first surface of the die **107**, and covering the surface of the portion of the oxide layer laterally protruding from the die **107**. According to the embodiments of the present disclosure, after the dies are transferred, the oxide layer included in the die bonded structure **200** protects the second surface (or referred to as the backside) of the die **107**, and the protective layer **120** included in the die bonded structure **200** protects the side surfaces and the first surface (or referred to as the front surface) of the die **107**. Therefore, the die bonded structure **200** of the embodiments of the present disclosure prevents the moisture and oxygen in the external environment from penetrating into the die **107**, and the reliability of electronic products is improved.

(33) FIG. 5 shows schematic cross-sectional diagrams of structures at some stages of a method of transferring dies and a die bonded structure after the dies are transferred according to another embodiment of the present disclosure. Referring to FIG. 5, first, an SOI substrate **101** is provided, and a die **107** and a bonding feature **109** are formed on the oxide layer **104** of the SOI substrate **101** and are separated from each other. The details the structure may refer to the aforementioned description of the FIG. 4, and will not be repeated here. Next, as shown in FIG. 5, at step **S501**, a supporting wafer **111** is provided, and spacers **113** are formed on the supporting wafer **111**. Then, a bonding process **510** is performed to bond the supporting wafer **111** to the bonding pads **117** of the bonding feature **109** through the spacers **113**, where the supporting wafer **111** is separated from the die **107** and the bonding feature **109** by a gap. In this embodiment, the bonding process **510** may be performed through the glue of the spacer **113**.

(34) Thereafter, as shown in FIG. 5, at step **S503**, the structure of the step **S501** is turned upside down, and the semiconductor substrate **102** of the SOI substrate **110** may be completely removed to expose the oxide layer **104**. Next, as shown in FIG. 5, at step **S505**, an etching process **520** is performed on the exposed oxide layer **104** and the protective layer **120** thereunder, so that the protective layer **120** and the oxide layer **104** in the die setting region **101A** of one repeating unit are separated from the protective layer **120** and the oxide layer **104** in the peripheral region **101B** of another adjacent repeating unit to form an oxide layer supporting feature **101**, and the structure

100-1 of transferring dies of FIG. **1** is completed. Afterwards, the process steps **S407**, **S409** and **S411** as shown in FIG. **4** are performed, using the transfer equipment **330** to release a large number of dies **107** from the structure **100-1** of transferring dies and to transfer the dies **107** onto the carrier substrate **201** to complete the fabrication of the die bonded structure **200**.

(35) The difference between the embodiments of FIG. **5** and FIG. **4** is that in the method of transferring dies of the embodiment of FIG. **5**, the etching process **520** of the oxide layer **104** is performed after the bonding process **510** of the supporting wafer **111** and after the removal of the semiconductor substrate **102**. In the method of transferring dies of the embodiment of FIG. **4**, the etching process **410** of the oxide layer **104** is performed before the bonding process **420** of the supporting wafer **111**.

(36) FIG. **6** shows schematic cross-sectional diagrams of structures at several stages of a method of transferring dies and a die bonded structure after the dies are transferred according to another embodiment of the present disclosure. Referring to FIG. **6**, first, an SOI substrate **101** is provided, and a die **107** and a bonding feature **109** are formed on the oxide layer **104** of the SOI substrate **101** and are separated from each other. The details of the structure may refer to the aforementioned description of FIG. **4**, and will not be repeated here. Next, referring to FIG. **6**, at step **S601**, an etching process **610** is performed on the oxide layer **104** and the protective layer **120** to form an oxide layer supporting feature **101**. The details of the structure may refer to the aforementioned description of the step **S401** of FIG. **4**, and will not be repeated here.

(37) Then, as shown in FIG. **6**, at step **S603**, a supporting wafer **111** is provided, and the supporting wafer **111** is etched by an etching process to form a cavity **112** and a stand-off of a spacer **114**. Then, an eutectic bonding bump is formed on the stand-off of the spacer **114** for bonding the supporting wafer **111** and the spacer **114** to the bonding pads **117** of the bonding feature **109** through a bonding process **620**, where the position of the die **107** corresponding to the cavity **112** of the supporting wafer **111** provides a larger space for accommodating the die **107**. In this embodiment, the bonding process **620** may be performed by eutectic bonding through the eutectic bonding bump of the spacer **114**.

(38) Next, as shown in FIG. **6**, at step **S605**, the structure of the step **S603** is turned upside down, and the semiconductor substrate **102** of the SOI substrate **110** may be completely removed to expose the oxide layer supporting feature **101**. Then, the structure **100-2** of transferring dies as shown in FIG. **1** is formed.

(39) Thereafter, as shown in steps **S607**, **S609** and **S611** of FIG. **6**, a large number of dies **107** are released from the structure **100-2** of transferring dies by using the transfer equipment **330**. Then, the dies are transferred onto the carrier substrate **201** to complete the die bonded structure **200** as shown in FIG. **6**. The details of the steps **S607**, **S609** and **S611** of FIG. **6** may refer to the aforementioned descriptions of the steps **S407**, **S409** and **S411** of FIG. **4**, and will not be repeated here.

(40) The difference between the embodiments of FIG. **6** and FIG. **5** is that in the method of transferring dies of the embodiment of FIG. **6**, the cavity **112** and the stand-off of the spacer **114** are formed in the supporting wafer **111**, and the bonding process is performed through the eutectic bonding bump of the spacer **114**.

(41) FIG. **7** shows schematic cross-sectional diagrams of structures at some stages of a method of transferring dies according to another embodiment of the present disclosure. Referring to FIG. **7**, first, an SOI substrate **101** is provided, and a die **107** and a bonding feature **109** are formed on the oxide layer **104** of the SOI substrate **101** and are separated from each other. The details of the structure may refer to the aforementioned description of FIG. **4**, and will not be repeated here. Next, as shown in FIG. **7**, at step **S701**, a supporting wafer **111** is provided, and the supporting wafer **111** is etched by an etching process to form a cavity **112** and a stand-off of a spacer **114**. Then, an eutectic bonding bump is formed on the stand-off of the spacer **114** for bonding the supporting wafer **111** and the spacer **114** to the bonding pads **117** of the bonding feature **109**

through a bonding process **710**, where the position of the die **107** corresponding to the cavity **112** of the supporting wafer **111** provides a larger space for accommodating the die **107**. In this embodiment, the bonding process **710** may be performed by eutectic bonding through the eutectic bonding bump of the spacer **114**.

(42) Then, as shown in FIG. 7, at step **S703**, the structure of the step **S701** is turned upside down, and the semiconductor substrate **102** of the SOI substrate **110** may be completely removed to expose the oxide layer **104** of the SOI substrate **110**. Next, as shown in FIG. 7, at step **S705**, an etching process **720** is performed on the exposed oxide layer **104** and the protective layer **120** thereunder, so that the protective layer **120** and the oxide layer **104** in the die setting region **101A** of one repeating unit are separated from the protective layer **120** and the oxide layer **104** in the peripheral region **101B** of another adjacent repeating unit to form an oxide layer supporting feature **101**. Then, the structure **100-2** of transferring die as shown in FIG. 1 is completed. Afterwards, the process steps shown in the steps **S607**, **S609** and **S611** of FIG. 6 are performed by using the transfer equipment **330** to release a large number of dies **107** from the structure **100-2** of transferring die, and to transfer the dies onto the carrier substrate **201**. Then, the fabrication of the die bonded structure **200** is completed.

(43) The difference between the embodiments of FIG. 7 and FIG. 6 is that in the method of transferring dies of the embodiment of FIG. 7, the etching process **720** of the oxide layer **104** is performed after the bonding process **710** of the supporting wafer **111** and after the removal of the semiconductor substrate **102**. In the method of transferring dies of the embodiment of FIG. 6, the etching process **610** of the oxide layer **104** is performed before the bonding process **620** of the supporting wafer **111**.

(44) FIG. 8 shows schematic cross-sectional diagrams of structures at some stages of a method of transferring dies and a die bonded structure after the dies are transferred according to another embodiment of the present disclosure. Referring to FIG. 8, first, according to some embodiments, an SOI substrate **101** is provided, and a die **107** and a bonding feature **109** are formed on the oxide layer **104** of the SOI substrate **101** and are separated from each other. The details of the structure may refer to the aforementioned description of FIG. 4, and will not be repeated here. The structure of the embodiment of FIG. 8 may be used to replace the structures of the embodiments of FIG. 4, FIG. 5, FIG. 6, and FIG. 7 to complete the structures **100-3** and **100-4** of transferring dies as shown in FIG. 2. The difference between the embodiment of FIG. 8 and the embodiments of FIG. 4, FIG. 5, FIG. 6, and FIG. 7 is that there are an under bump metallurgy (UBM) **122** and a metal bump **124** disposed on the conductive pad **115** of the die **107** in the embodiment of FIG. 8. Next, the process steps as shown in the steps **S401**, **S403** and **S405** of FIG. 4, or as shown in the steps **S501**, **S503** and **S505** of FIG. 5, or as shown in the steps **S601**, **S603** and **S605** of FIG. 6, or as shown in the steps **S701**, **S703** and **S705** of FIG. 7 are performed to form the structure **100-3** or **100-4** of transferring dies as shown in FIG. 2.

(45) Thereafter, as shown in steps **S801**, **S803** and **S805** of FIG. 8, in some embodiments, a large number of dies **107** are released from the structure **100-3** of transferring dies by using the transfer equipment **330**, and then the dies are transferred onto the carrier substrate **201** to complete the die bonded structure **200** as shown in FIG. 8. Similarly, a large number of dies **107** in the structure **100-4** of transferring dies as shown in FIG. 2 may be transferred by the process steps **S801**, **S803** and **S805** as shown in FIG. 8 to complete the die bonded structure **200** as shown in FIG. 8.

(46) FIG. 9 is a top view of a die bonded structure **200** after the dies are transferred according to some embodiments of the present disclosure. In order to make the diagram be concise and easy to understand, FIG. 9 only depicts an oxide layer of the die bonded structure **200**, such as the die setting region **101A** of the oxide layer supporting feature **101**, a carrier substrate **201**, and multiple wires **203** on the carrier substrate **201**, where the position of the die setting regions **101A** of the oxide layer supporting feature **101** is the position of the transferred dies. These dies are electrically connected to the wires **203** on the carrier substrate **201**, respectively. In some embodiments, the

wires **203** are, for example, signal wires, and the signal of the dies may be transmitted to other devices of an electronic product, or the signal of a peripheral circuit of the electronic product may be transmitted to the dies through the wires **203**.

(47) According to the embodiments of the present disclosure, in a mass transfer of dies, the number of dies per unit area is increased, thereby improving the efficiency of transferring dies. Meanwhile, in the die bonded structure after the dies are transferred, the front and side surfaces of the dies are covered by a protective layer and the back surface of the dies is also covered by an oxide layer to prevent moisture and oxygen from penetrating into the dies, thereby improving the reliability of electronic products using these dies.

(48) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A structure of transferring dies, comprising: an oxide layer supporting feature, including a plurality of repeating units, wherein each of the repeating units includes a die setting region and a peripheral region, and the die setting region of one repeating unit is separated from the peripheral region of another adjacent repeating unit; a plurality of dies, respectively disposed on the die setting regions but not disposed on the peripheral region of the oxide layer supporting feature; a bonding feature, disposed on the peripheral region of the oxide layer supporting feature; a supporting wafer, disposed under the oxide layer supporting feature and separated from the dies and the bonding feature by a gap; and a spacer, disposed between the bonding feature and the supporting wafer, wherein the spacer is bonded to the bonding feature, and comprises a glue bond dam or an eutectic bond bump.
 2. The structure of transferring dies of claim 1, wherein the bonding feature is separated from the die, and the bonding feature and the die have a same semiconductor layer and a same interconnect structure layer.
 3. The structure of transferring dies of claim 1, wherein the bonding feature comprises a bonding pad, and the glue bond dam or the eutectic bond bump is bonded to the bonding pad.
 4. The structure of transferring dies of claim 1, wherein the die comprises a plurality of conductive pads, or the die comprises a plurality of conductive pads, and an under bump metallurgy and a metal bump disposed on the conductive pad.
 5. The structure of transferring dies of claim 1, wherein the supporting wafer comprises a cavity corresponding to the die, and the distance between the die and a bottom surface of the cavity of the supporting wafer is greater than the distance between the bonding feature and a surface of the supporting wafer.
 6. The structure of transferring dies of claim 1, wherein a thinned configuration is provided at an interface between the peripheral region and the die setting region, and the peripheral region is separated from the die setting region of the same repeating unit of the oxide layer supporting feature when transferring the dies.
 7. The structure of transferring dies of claim 1, further comprising a protective layer covering the dies, the bonding feature, and a surface of the oxide layer supporting feature exposed between the die and the bonding feature, wherein the protective layer includes a plurality of openings to expose a plurality of conductive pads or a plurality of metal bumps of the dies, and a bonding pad of the bonding feature.
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